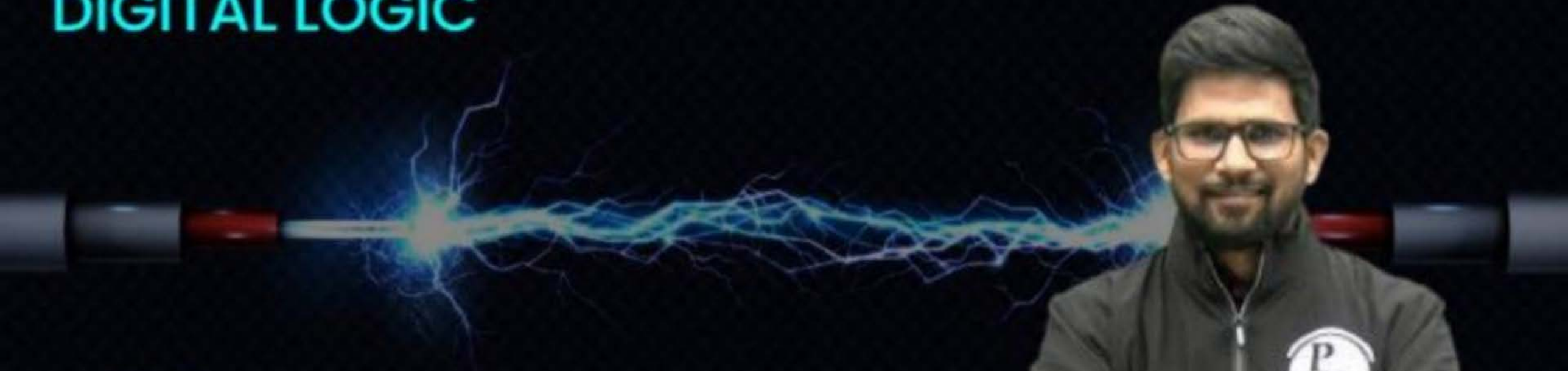
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 05

Combinational Circuit



By- Chandan Gupta Sir


Recap of
Previous
Lecture


H.S., F.S.



- F.S. Cont.
- K-Map.

H.W.

Q.



$y_1 = x \oplus y \oplus \bar{x}y = \overline{\bar{x} \oplus y \oplus \bar{x}y}$

$= \overline{\bar{x}+y} = x \cdot \bar{y}$

$y_2 = \overline{x \oplus y} \cdot \bar{x}y = (\bar{x}\bar{y} + xy)\,\bar{x}y = 0$

Q. $A = (1010)_2$

$B = (0010)_2$

perform $(A-B)$ & $(B-A)$ and find out result $\Longrightarrow (A-B) = (01000)_2$, $(B-A) = (11000)_2$

Q. $A = (0111)_2$ What is the result of $(A-B)$?

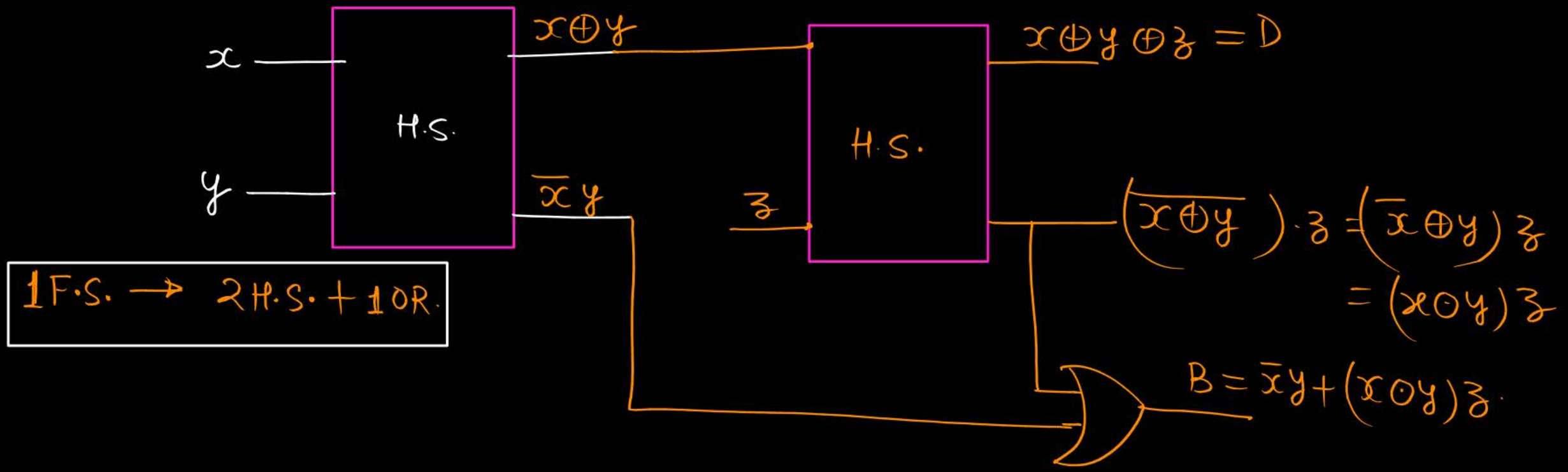
$B = (1011)_2$

$(A-B) = (11100)_2$

- Implementation using H.S and OR gate :

x, y, z $(x\text{-}y\text{-}z)$ $\longrightarrow$ $D = x \oplus y \oplus z = \Sigma(1,2,4,7)$

$B = \bar{x}y + yz + \bar{x}z = \bar{x}y + (\overline{x \oplus y})z$



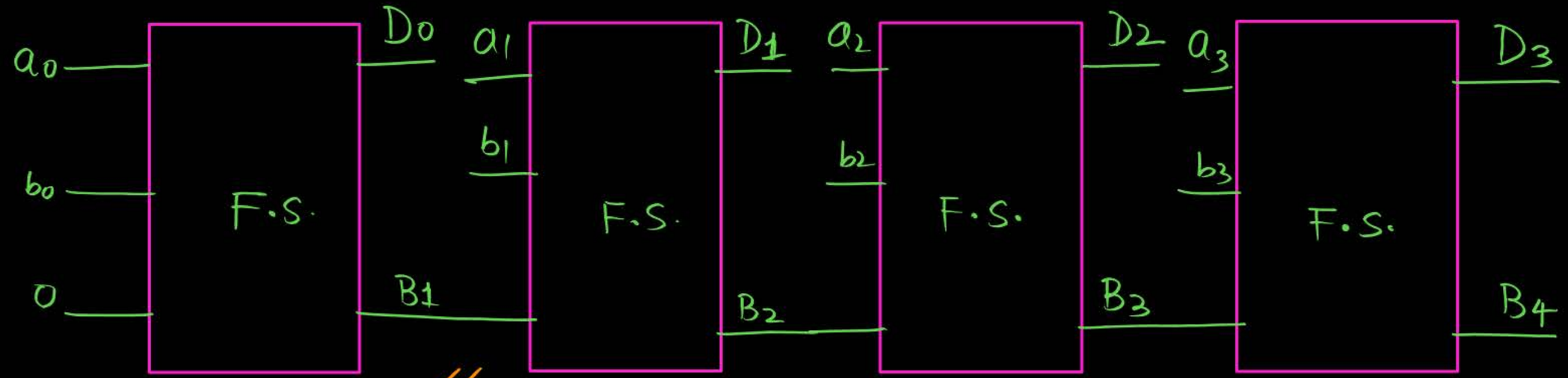
- $A = a_3\ a_2\ a_1\ a_0$

— $B = b_3\ b_2\ b_1\ b_0$

$B_3\ B_2\ B_1\ 0$

Result $B_4\ D_3\ D_2\ D_1\ D_0$

- To subtract two n-bit numbers,, number of H.S. and number of OR gate required is :



Result = $B_4\ D_3\ D_2\ D_1\ D_0$

$1\ H.S. + (n-1)\ F.S.$

$1\ H.S. + 2(n-1)\ H.S. + (n-1)\ OR$

$\boxed{(2n-1)\ H.S. + (n-1)\ OR}$

Question

Two subtract two 4-bit numbers, minimum number of H.S. required is m and minimum number of OR gates require is n then value of $(m + n)$ is 10.

1 H.S. + 3 F.S.

1 H.S. + 6 H.S + 3 OR

= 7 H.S. + 3 OR gate

$m = 7$

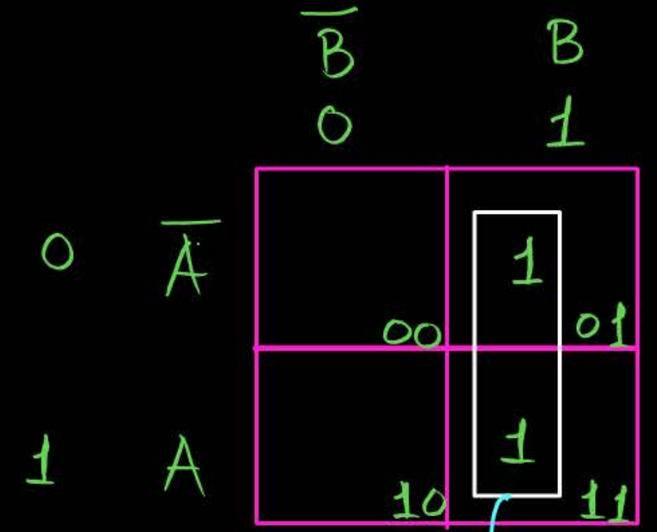
$n = 3$

K-Map

Used for Entries :

- '0'
- '1'
- 'X'
- Variable entry.

$f(A,B) = \Sigma(1,3) = B$

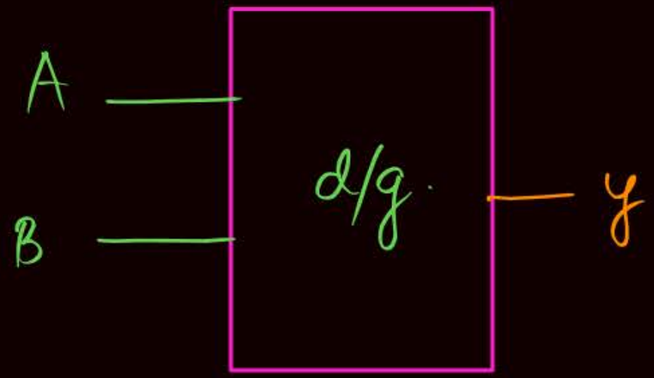


(B)

$= \bar{A}B + AB$

$= B[\bar{A} + A]$

$= B \cdot 1 = B$



A = 0 & B = 1 ⟶ not possible

if both switch ON or both switch off O/P = 1 and '0' otherwise.

A	B	y
0	0	1
0	1	X
1	0	0
1	1	1

$$y(A,B) = \Sigma(0,3) + d\,\Sigma(1)$$

BCD Code

	2^3	2^2	2^1	2^0
	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

1010
1011
1100
1101
1110
1111

→ invalid BCD Codes

$(23)_{10} = (10111)_2 = (0010\ 0011)_{BCD}$

$(157)_{10} = (\quad)_2$

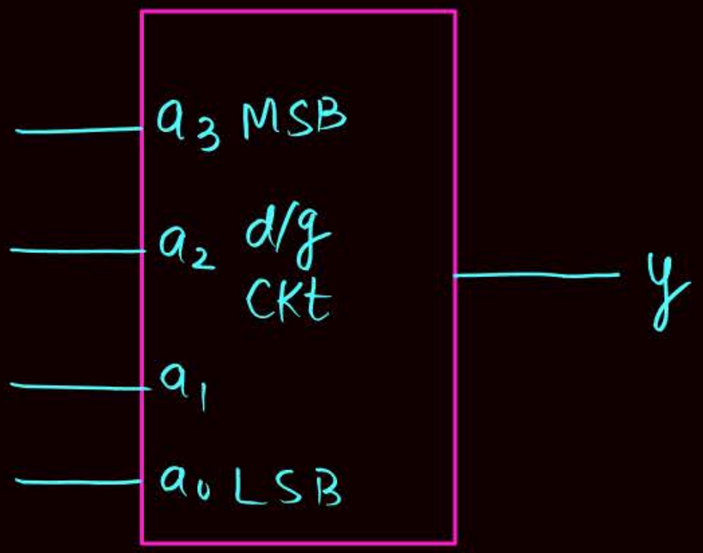
$= (0001\ 0101\ 0111)_{BCD}$

$(10)_{10} = (0001\ 0000)_{BCD}$

Design a CKt which takes BCD code as i/p and generate o/p '1' if i/p BCD code is divisible by either 2 or 3.

i/p-lines :→ 4-i/p lines

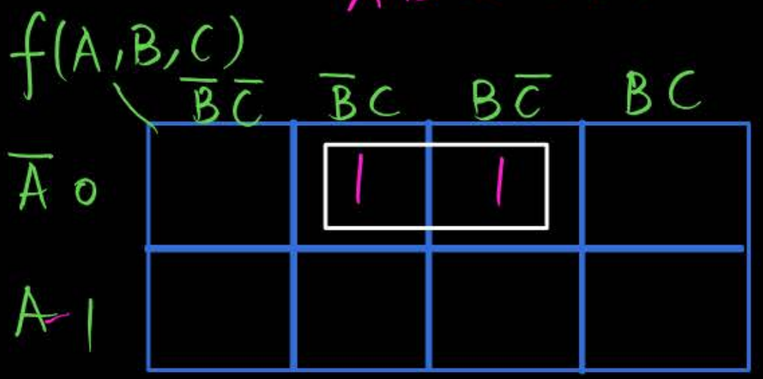
o/p-lines → 1 y.

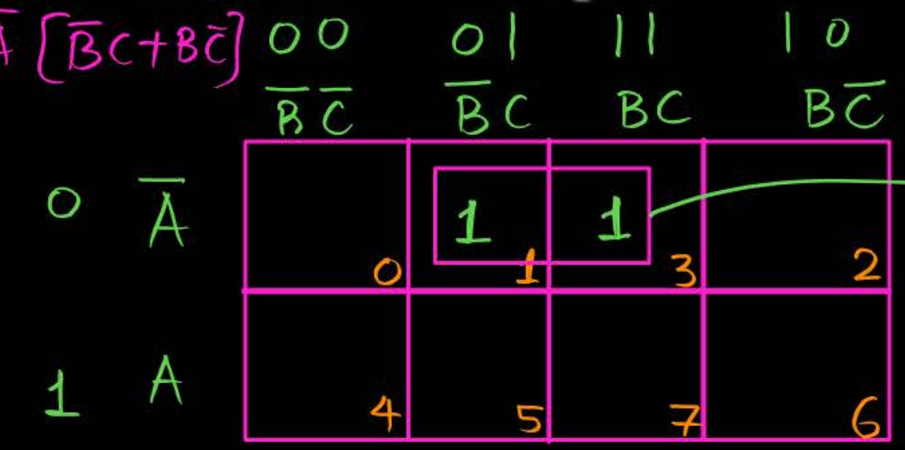


a_3	a_2	a_1	a_0	y_1
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

- Why gray code is used in case of K-map ?

$\bar{A}\bar{B}C + \bar{A}B\bar{C} = \bar{A}[\bar{B}C + B\bar{C}]$





$\bar{A}C$

$\bar{A}\bar{B}C + \bar{A}BC$

$= \bar{A}C[\bar{B}+B] = \bar{A}C$

$0 \rightarrow (00)_2 \rightarrow (00)_{gray}$

$1 \rightarrow (01)_2 \rightarrow (01)_{gray}$

$2 \rightarrow (10)_2 \rightarrow (11)_{gray}$

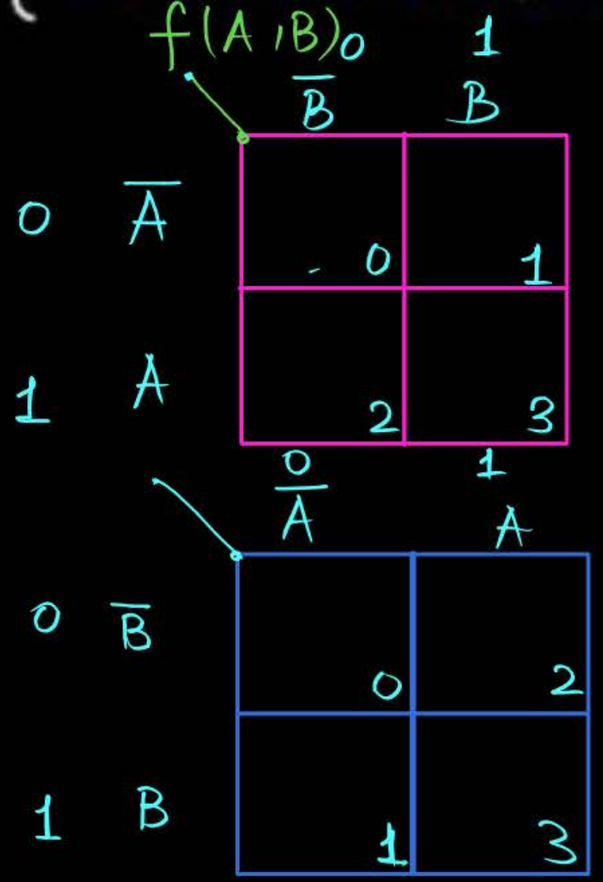
$3 \rightarrow (11)_2 \rightarrow (10)_{gray}$

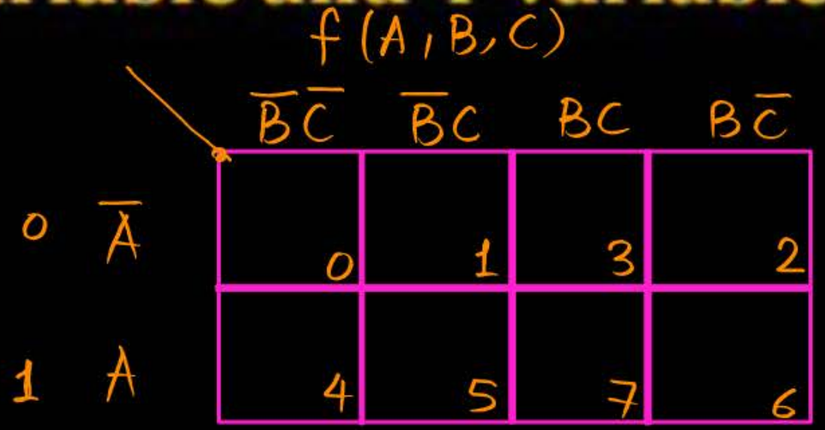
$(23)_{10} \rightarrow (10111)_2 \rightarrow (11100)_{gray}$

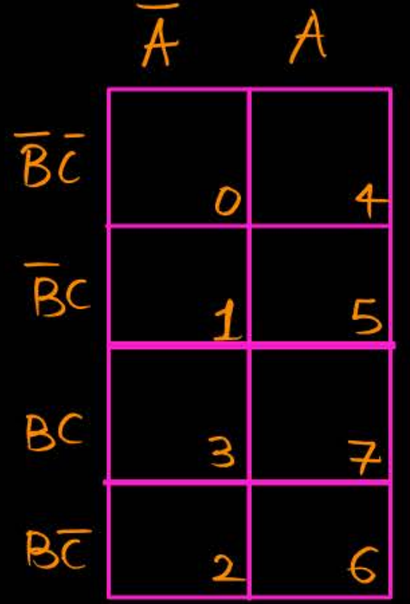
$(24)_{10} \rightarrow (11000)_2 \rightarrow (10100)_{gray}$

$(25)_{10} \rightarrow (11001)_2 \rightarrow (10101)_{gray}$

2-variable, 3-variable and 4-variable K-maps







$f(A,B,C,D)$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
AB	12	13	15	14
$A\bar{B}$	8	9	11	10

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	0	4	12	8
$\bar{C}D$	1	5	13	9
CD	3	7	15	11
$C\bar{D}$	2	6	14	10



2 Minute Summary

- F.S.
- Basics of K Map.
- BCD code.

Thank you

GW

Soldiers !

